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Day reference map



Night query frame

Goal: recognize the robot's location at night using a daytime map

Robotics need

GPS may be unreliable or unavailable indoors and around buildings. Visual place recognition offers a way to localize from camera data alone.

Main challenge

At night, illumination changes remove texture and color cues, so descriptors trained mostly on daytime imagery can fail.

Applied CV angle

We want not only benchmark performance, but also a live system that can support webcam, phone, and robot-camera localization.

Research question

Which visual descriptor is most effective for day-to-night place recognition in our setting, and can a strong descriptor backbone support both benchmark evaluation and a practical live localization demo?

Main hypothesis

The choice of descriptor is the dominant factor in our current nocturnal VPR pipeline, and modern VPR-specific descriptors should outperform older or more generic baselines on both controlled benchmarks and our campus dataset.

Focus of this phase: descriptor exploration

Training-free evaluation

Classical / older baselines

- SAD
- AlexNet conv3
- NetVLAD

Modern VPR descriptors

- CosPlace
- EigenPlaces
- SALAD
- PatchNetVLAD

Night / temporal directions we reviewed

- SeqSLAM
- NocPlace
- VPRTempo
- lightweight re-ranking / temporal consistency

Why SALAD stood out

CVPR 2024 descriptor aggregation method with strong large-scale VPR results using a DINOv2 backbone.

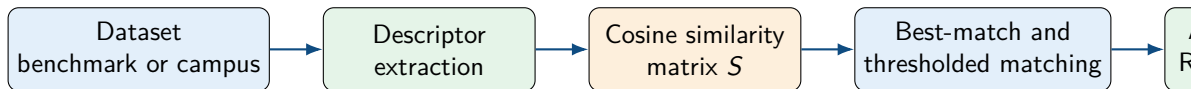
Descriptor literature



Night / temporal work



Our choice:
test strong
descriptors first



Controlled evaluation

GardensPoint remains the main benchmark because it directly tests day-to-night place recognition.

Applied validation

The campus datasets let us study transfer, multi-view effects, and deployment-oriented behavior.

GardensPoint

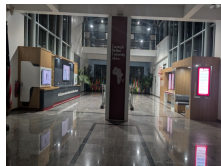
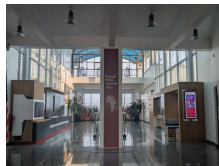
- 200 day-right reference images
- 200 night-right query images
- controlled day-to-night benchmark
- primary benchmark for this project

Why not Pittsburgh as the main benchmark?

Pittsburgh is useful for general VPR, but it is less aligned with our day-to-night problem statement.

Campus dataset

- 50 day reference images
- 64 night query images
- original strict protocol: 33 matched, 31 no-match
- place-level protocol: 64 matched queries over 10 place IDs



Baseline descriptors

- SAD
- AlexNet conv3
- CosPlace
- EigenPlaces

Additional descriptors

- NetVLAD
- PatchNetVLAD
- HDC-DELF
- SALAD

Planned but not main focus here

- CLAHE preprocessing
- sequence matching
- VPRTempo exploration
- live search optimization

This presentation phase focuses on descriptor exploration first

Descriptor	AUC	R@1	R@5	R@10
CosPlace	0.592	0.350	0.840	0.925
EigenPlaces	0.676	0.370	0.840	0.940
AlexNet conv3	0.195	0.300	0.665	0.745
SAD	0.030	0.090	0.225	0.305

Interpretation

The early baseline story was that modern place descriptors were already putting the correct place into the top few candidates, but ranking the correct match first was still difficult.

Metric	EigenPlaces	SALAD
AUC	0.676	0.924
R@100P	—	0.385
R@1	0.370	0.450
R@5	0.840	0.955
R@10	0.940	0.990

Observed improvement

SALAD gave the strongest controlled benchmark result we have tested so far.

Examples: 1 correct (green) and 1 wrong (red) matches from $S \geq \text{thresh}$



Why we should be skeptical

The SALAD benchmark result is much stronger than the earlier baselines, so we reviewed the code path as if we were strict examiners.

What looks correct

- separate day and night folders
- no filename leakage into inference
- normalized descriptors + cosine similarity
- Recall@K uses exact hard ground truth

What should be interpreted carefully

- GardensPoint soft ground truth allows nearby frames
- AUC and R@100P are therefore more lenient
- strongest evidence is really R@1 / R@5 / R@10

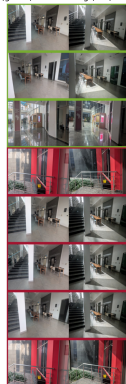
Conclusion: strong and plausible result, but top-k is the most defensible evidence

Descriptor	AUC	R@100P	R@1	R@5	R@10
CosPlace	0.470	0.030	0.727	1.000	1.000
EigenPlaces	0.443	0.000	0.788	1.000	1.000
AlexNet	0.512	0.061	0.758	0.939	0.939
HDC-DELF	0.488	0.030	0.788	0.939	0.939
SALAD	0.504	0.030	0.818	1.000	1.000

Interpretation

Under strict 1-to-1 labeling, SALAD gives the best top-1 retrieval, but open-set rejection remains difficult.

Examples: 3 correct (green) and 5 wrong (red) matches from $S > \text{thresh}$





Main lesson

The original campus evaluation was too strict for a real multi-view place recognition task. Several same-place but different-view matches were being counted as wrong.

What changed

We created a duplicate place-level dataset:

- 10 place IDs
- 64 matched queries
- 12 queries rematched from the old `-npm` set

Descriptor	AUC	R@100P	R@1	R@5
CosPlace	0.627	0.114	0.969	1.000
EigenPlaces	0.663	0.110	0.922	1.000
PatchNetVLAD	0.641	0.074	0.938	1.000
HDC-DELF	0.507	0.132	0.922	0.969
SALAD	0.653	0.130	0.969	1.000

Interpretation

Once valid alternate views are credited properly, SALAD becomes one of the strongest campus descriptors overall.

Examples: 3 correct (green) and 5 wrong (red) matches from $S \geq \text{thresh}$



Best controlled benchmark

SALAD

strongest GardensPoint result

Best overall place-level AUC

EigenPlaces

still extremely competitive

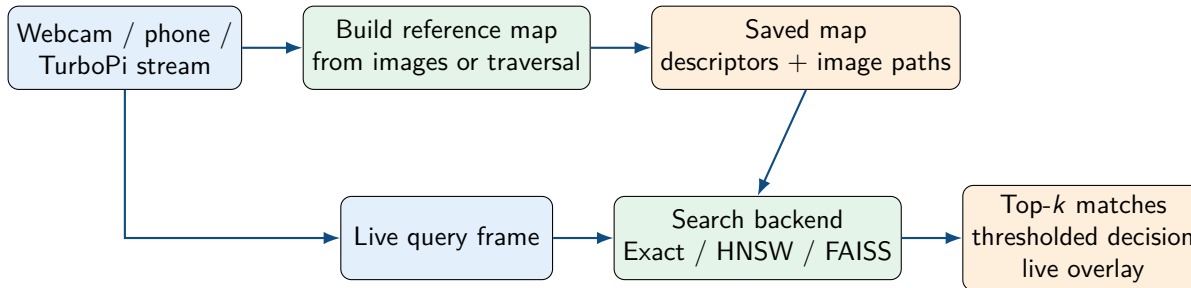
Best place-level top-1

CosPlace / SALAD tie

both reach 0.969

Takeaway

Descriptor choice matters a lot. Modern descriptors clearly outperform the older baselines, and SALAD gives the strongest benchmark result we have observed so far.



Modular config-driven pipeline

Map build + live localization

Robot-facing demo path

CLAHE

Contrast enhancement to reduce the day-night appearance gap before descriptor extraction.

Sequence matching

Temporal consistency to reinforce nearby frames and improve ranking stability.

Future descriptor work

Continue with VPRTempo, BoQ, MixVPR, AnyLoc, and other modern extractors.

Current decision

For this phase, we focused on descriptor exploration first so that we can establish a strong backbone before adding more pipeline complexity.

- ① Our hypothesis was supported: descriptor choice has a major effect on nocturnal VPR performance.
- ② GardensPoint remains the right primary benchmark because it directly matches the day-to-night problem.
- ③ SALAD is the strongest benchmark descriptor we have tested so far, but its top- k gains are more trustworthy than the threshold-based metrics alone.
- ④ The campus relabeling showed that the original strict protocol under-credited valid multi-view same-place matches.
- ⑤ The live system is already in place, so the next phase is to connect stronger descriptors and lightweight temporal improvements to the demo path.

Presentation

Sunday, April 26, 2026

2:00 PM, F203

8 minutes presentation

4 minutes Q&A

Random presentation order

Required deliverables

5-minute video presentation

Due Saturday, April 25, 11:59 PM

Final PDF report

Reproducible GitHub link

Video demo link (if applicable)

Thank you

Demo / Questions